

Small-Scale Text Diffusion for a Psychology Assistant: A Discrete-Diffusion Pipeline, a Blind-Annotation Retraction, and an Honest Data-Bound Scaling Null

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June 2026

Working preprint. Single-GPU (RTX 3090) study; all artifacts in the project repository.

Abstract

We build an English psychology / mental-health text-generation model on a single consumer GPU and report the full chain of results — several of them negative — with deliberate measurement rigor. The methodological spine is a self-caught error: an effect we first “found” non-blind (corpus cleaning halving confabulation, $57\% \rightarrow 29\%$) *vanished under blind annotation* (Fisher $p = 1.0$); we retract it and treat the caught confirmation bias, and the blind protocol that exposed it, as a primary contribution. Substantive findings: (1) **Architecture** — continuous (Gaussian) text diffusion at 60.8M parameters / 2.56B tokens produces topic-coherent *word salad*, while discrete masked diffusion (MDLM) on the same data, tokenizer, and token budget — at a marginally *larger* 66.8M, a gap the scale-axis null (§9) shows is immaterial — yields grammatical sentences. (2) **Decoding** — the discrete model’s dominant failure, repetition collapse, is a *decode artifact, not a capacity limit*: confidence-based unmasking plus repetition controls remove it with no weight change (rep-1 -53% at fixed temperature). (3) **Data** — a strict-filtered 0.77B-token corpus is $3.3\times$ cheaper than the 2.56B noisy corpus and no worse on the most sensitive eval (paired cosine $d = +0.0094$, win-rate 0.527): cleaning buys *cost*, not quality. (4) **Scale** — doubling to 120M parameters at fixed data moves neither aggregate quality (paired $d = -0.0094$, win-rate 0.479) nor the blind homonym-drift floor (Fisher $p = 0.77$); a measured null we attribute to an *inferred* data-bound regime (6.4 vs. 11.6 tokens/param against a ≈ 20 optimum), falsifiable only by a data-fed rerun. The shipped model is the cheaper 66.8M clean checkpoint, deployed fully on-device (Android, ONNX, fp32). A closing audit re-derived every headline statistic from raw artifacts (Appendix A).

Intended use, limitations, and safety. This is a research artifact for studying small-scale text diffusion, *not* a clinical, diagnostic, or psychological tool. The model was validated on no clinical endpoint, carries no regulatory clearance, and its outputs are frequently vague, repetitive, and factually wrong — a blind homonym-drift floor of 25–32% persists at both model sizes (§9). It must not be used for diagnosis, treatment, or crisis support, and the on-device demo (§10) is an engineering proof-of-concept, not a product. For crisis support contact 988 (US) or a local emergency line. The training data contain no real conversations and no personally identifying information.

Code, model, and artifacts. Code, evaluation artifacts (blind keys, annotator labels, eval harness), and the L^AT_EX source of this paper: <https://github.com/thunder-phoenix-lab/research-psyche-diffuse-66m>. Model weights (66.8M production checkpoint + 120M scale checkpoint, SentencePiece tokenizer, ONNX mobile export): <https://huggingface.co/tp-di/psyche-diffuse-66m>. Every headline number recomputes bit-exactly from the shipped artifacts (Appendix A).

1 Introduction

The goal was a compact diffusion language model specialized to psychology and mental-health dialogue, trainable and deployable on one RTX 3090. Diffusion LMs promise controllable, non-autoregressive generation, but little is documented about what works at the small scale a hobbyist or edge budget allows. This paper is an engineering-and-science log: what failed, what worked, and where our own measurements misled us until corrected.

The project ran as seven stages. For each, the body states *what was done*, *why*, and *what it led to* — the causal chain matters, because three pivots (architecture switch, sampler rewrite, corpus rebuild) were each forced by a documented negative result, and one headline “positive” was retracted on its own evidence:

1. **Data + tokenizer (§2)**: 2.56B-token psychology-dominant corpus, 261k-pair SFT set, fixed 16k SentencePiece vocab → one tokenizer and recipe frozen for every later arm.
2. **Continuous diffusion, v1–v3 (§3)**: topic control emerged; grammar never did; five extra sampler tricks made outputs strictly worse → pivot to discrete diffusion at identical budget.
3. **Discrete MDLM (§4)**: same scale, grammatical English → pipeline gate opened.
4. **Decode fix (§5)**: repetition collapse removed with zero retraining → collapse = decode order, sampler frozen for all evals and deployment.
5. **Clean corpus + factorial (§6–7)**: strict filter to 0.774B tokens; non-blind “confab halved” *retracted* under blind labels → standing blind protocol.
6. **Data and scale axes (§8–9)**: paired 1000-prompt eval + blind traps → cleaning buys cost not quality; 2× params at fixed data is a null.
7. **Deployment (§10)**: 66.8M-clean exported to ONNX, runs on-device.

Related work. The discrete model follows masked-diffusion LMs — MDLM [1], SEDD [2] — with confidence-ordered unmasking from MaskGIT [3]; the continuous baseline is Diffusion-LM [4] with DiffuSeq-style conditioned SFT [5]; the backbone is a DiT [9]. Scaling is read through Chinchilla [6]; the data result confirms, in the diffusion regime, the curated-small-data line of TinyStories [7] and phi [8]. The combination — masked diffusion, single GPU, domain-specialized, with a blind-annotation retraction — is, to our knowledge, not covered by these.

2 Setup: model, objective, data

Backbone. Diffusion Transformer with AdaLN-zero timestep conditioning, RMSNorm, SwiGLU MLP, SDPA attention, learned positions, tied output head ($\text{logits} = hE^\top + b$), sequence length 512. Two sizes (Table 1); 120M scales width and depth at fixed head dim 64. Training is pure bf16 — parameters, activations, gradients; fp32 only inside RMSNorm, the sinusoidal time embedding, and

loss reductions — with AdamW8bit [10], no fp32 master weights. This single-card regime is held constant across all arms, so it is not a confound for any comparison.

Table 1: Architectures. Parameter counts are from training-log prints, not estimates.

name	hidden	layers	heads	MLP inter	params	tok/param @0.774B
66.8M	512	12	8	1408	66.8M	11.6
120M	640	15	10	1664	121.4M	6.4

Objective. MDLM [1]: per sequence draw $t \sim U(0.01, 0.99)$, mask each token independently with probability t (model vocab 16,001: the 16,000 SentencePiece pieces plus the corruption symbol [MASK] at id 16000), cross-entropy on masked positions only. SFT is DiffuSeq-style: only response positions are noised, prompts stay clean as conditioning; zero new parameters, pretrain checkpoint loaded `strict=True`.

Tokenizer. A 16,000-piece SentencePiece BPE was trained from scratch on the English pretrain corpus — byte-fallback (no out-of-vocabulary tokens), reserved `<pad>/<unk>/<s>/</s>` at ids 0–3 — rather than reusing an off-the-shelf vocabulary, so subword units track the clinical register (*psychotherapy, depression, cognitive, schizophrenia* each map to a single piece). 16k was chosen as a compact vocabulary for this parameter budget — the embedding table is material at 66.8M — and vocabulary size was not ablated. The [MASK] corruption symbol is one extra embedding row appended at id 16000 (model vocab 16001), not an SPM piece. The same frozen tokenizer is used by every arm including the continuous baseline (§3), so it confounds no reported contrast, and all token counts and the tok/param figures (Table 1) are in its units; the exact tokenizer ships as `spm_en.model` with the model release.

Recipe (frozen). Batch $32 \times \text{accum } 8$ ($= 256$ seqs $= 131\text{k}$ tokens/step), LR $4 \cdot 10^{-4}$ cosine, 2% warmup, weight decay 0.1, betas (0.9, 0.99), clip 1.0; pretrain 20k steps ($\approx 2.62\text{B}$ seen tokens $= \approx 1$ epoch noisy / ≈ 3.4 epochs clean), SFT 8k. Wall-clock per run on the 3090: $\approx 9.3\text{h}$ pretrain + 3.7–3.85h SFT at 66.8M; $\approx 26.5\text{h} + 5.8\text{h}$ at 120M (run ledger: Appendix D).

Data. Pretrain corpus, psychology-dominant English, two versions: **noisy** 2,559,832,146 tokens / 2,163,570 docs ($\approx 69\%$ loosely psych-filtered Pile, plus peS2o, OpenAlex psychology, PMC, Gutenberg, PsyArXiv, and a small RU→EN translated set); **clean** 773,925,936 tokens / 832,822 docs (§6). SFT: 261,443 (prompt, response) pairs at length 512 (assistant + mental-health + translated psychology dialogue), shared by every arm. We do *not* redistribute the corpora — the noisy mix is $\approx 69\%$ Pile-derived and the SFT mix has heterogeneous source licenses — but release the exact filter/build scripts and token/doc counts so both versions are reconstructible; the 1000-question eval set ships as prompts, with gold answers rebuildable from the source datasets.

What it led to. Every later contrast changes one factor on identical infrastructure. The one violation of this rule was the non-blind labeling episode (§7) — caught and retracted.

3 Stage 1 — continuous diffusion fails, three samplers deep

We first built a continuous, x_0 -parameterized Diffusion-LM-style model: 60.8M params, 2.56B tokens, DDIM, classifier-free guidance, self-conditioning.

v1–v2. CFG controls topic: at $\text{cfg}\approx 2.5$ outputs lock into a therapy register (on-topic content-word density $\sim 10\% \rightarrow \sim 35\%$), and constraints/clamp suppress garbage. Grammar never emerges — outputs are word salad in the right vocabulary; CBT-vs-ACT and Vygotsky prompts fail outright. **v3 (negative).** Five further techniques (CFG cosine ramp, stochastic DDIM, final top- p /temp, iterative refinement, short max-length) were ablated on the same checkpoint: together they made outputs strictly worse — register lock-in $3/5 \rightarrow 0/5$, on-topic density $\sim 35\% \rightarrow \sim 15\%$; final top- p over 16k SPM occasionally emits rare-Unicode one-shots.

What it led to. Inference-time engineering has a ceiling and we hit it: the limit is architectural. The arm survives only as a matched-budget baseline.

4 Stage 2 — discrete MDLM gives grammar at the same scale

The 66.8M pretrain reached masked-CE $9.80 \rightarrow 4.28$ (token acc 29.9%) in 20k steps; SFT $4.80 \rightarrow 3.71$ (35.8%). Coherent clauses appeared at 20% of training; post-SFT, coping/advice prompts produce grammatical multi-sentence on-topic English — the failure mode shifted to repetition collapse on factual/homonym prompts, expected at 60M.

What it led to. The architecture finding (discrete $\gg$ continuous at small scale, holding scale/data/tokenizer fixed), a frozen pipeline for matched experiments, and a defined next target: collapse.

5 Stage 3 — repetition collapse is decode-bound

Replacing random-order unmasking with confidence-ordered unmasking (cosine schedule) plus repetition penalty 1.3, neighbor-ban 2, no-repeat-3-gram fixes the same checkpoint, no retraining (Table 2).

Table 2: v1 vs. v2 sampler, identical 66.8M-noisy weights, 5 matched prompts, temp fixed 0.9.

metric	v1 random	v2 conf+rep	Δ
distinct-1	0.647	0.835	+29%
distinct-2	0.929	0.971	+0.042
rep-1	0.353	0.165	−53%
max-token-freq	0.090	0.064	−29%

A first draft compared v1 at temp $\{0.7, 1.0\}$ to v2 at 0.9 and showed rep-1 −62%; rerun at fixed 0.9 the effect attenuated to −53% — part of the original delta *was* temperature, and the controlled number is reported. **What it led to:** collapse = decode order, not capacity; sampler frozen for all evals (and the phone). The decode fix did *not* reduce homonym drift, isolating it as a data/capacity question.

6 Stage 4 — strict-filtered clean corpus

Drift such as *Vygotsky* $\rightarrow$ “coastal/coral” implicated the Pile component. Strict filter — keep iff core-psych hits ≥ 3 and core $\geq$ off-domain — yields 0.774B tokens (30% of noisy) at far higher purity;

the standing rule — “*never weaken the filter to buy tokens*” — held, accepting under-feeding of 120M. A pretrain-level probe (clean stays developmental on *proximal/distal*; noisy drifts to islands) motivated — but does not establish — a corpus effect; that test is §7. **Led to:** the corpus axis and the 3.3× cheaper production model.

7 Stage 5 — the retracted finding and the blind protocol

Factorial: decode (v1/v2) × corpus (noisy/clean) at 66.8M. Decode reproduced on both corpora. On 14 homonym traps × 2 seeds the author labeled drift *knowing condition and hypothesis*: 16/28 vs 8/28 — “cleaning halves confabulation”. Three checks before publishing: (1) **CI**s: Wilson [0.39, 0.73] vs [0.15, 0.47], Fisher $p = 0.058$ — not significant. (2) **Blind** (decisive): 56 pooled, shuffled, source-stripped items, labeled by two fresh, independent annotators — LLM (Claude) sub-agents with no memory of the project and no exposure to source, hypothesis, or condition (verbatim task spec in Appendix C). Using fresh LLM sub-agents rather than the author is what makes the blind protocol cheap enough to run as standing practice:

annotator	noisy	clean	Fisher p
author, non-blind	16/28 (57%)	8/28 (29%)	0.058
A, blind	11/28 (39%)	10/28 (36%)	1.000
B, blind	7/28 (25%)	7/28 (25%)	1.000

$\kappa_{AB} = 0.71$: criterion reliable, effect absent — the non-blind delta was confirmation bias. (3) **Specificity**: clean is not blander (TTR 0.899 vs 0.900; content-word 0.556 vs 0.562; hapax 0.910 vs 0.907).

Retraction; reading. Trap drift splits $\sim 14/16$ homonym-type (*drive*→vehicles) — corpus-invariant lexical floor — vs $\sim 2/16$ contamination-type, all cleaning can fix. **Led to:** standing protocol — pool, shuffle, strip, ≥ 2 fresh blind annotators, decode after, report κ /CI/Fisher — used throughout §9.

8 Stage 6a — data axis: cleaning buys cost, not quality

Same arch/SFT; pretrain clean 0.774B vs noisy 2.56B. Ceiling metrics: tie (above). Degeneration, 1000 prompts: distinct-2 0.993/0.991, rep-1 0.137/0.141, zero collapse. Sensitive eval, MiniLM cosine-to-gold on 725 paired prompts: 0.500 vs 0.490, $d = +0.0094$, CI [+0.0003, +0.0186], $t(724) = 2.01$, win 0.527 — borderline significant, negligible size. **Led to:** cleaning’s benefit is cost (3.3× fewer tokens, no penalty); confirms TinyStories/phi in diffusion. Caveats: volume/purity confounded (noisy-0.77B not run); slim fluency edge $\neq$ trap result (§7); the ± 0.0094 symmetry with §9 is coincidental (not engineered or tuned) — it shows the metric resolves 0.01, but the signs are single-seed.

9 Stage 6b — scale axis: a data-bound null

120M on identical clean data (transient CUDA fault absorbed by auto-resume). Pre-registered evals: **Aggregate paired**: 0.490 vs 0.500, $d = -0.0094$, CI [−0.0188, −0.0001], $t = -1.96$, win 0.479 — practical tie. **Blind traps** (3 fresh blind annotators — again independent Claude sub-agents, §C — κ 0.767/0.867/0.825): 120M 7/28 vs 66.8M 9/28, Newcombe [−0.294, +0.161], Fisher $p = 0.768$ — floor 25–32% both. **Led to:** measured null; inferred data-bound mechanism (6.4 vs 11.6 tok/param,

optimum ≈ 20); production = 66.8M-clean. Scope: single seed/cell, two checkpoints, in-distribution golds; falsifier = 2.4B-token 120M.

10 Stage 7 — on-device deployment; conclusions

66.8M-clean exported to ONNX (286 MiB = 300 MB asset; debug APK 365 MB), ONNX Runtime Mobile CPU, fp32 inference of bf16 weights (no quantization), v2 sampler ported, diffusion visualized (Figure 1).

Limitations. Every contrast is a single seed and a single checkpoint per cell on one GPU; we report paired tests and confidence intervals but no across-seed variance, so the small effects (§8–9) should be read as point estimates, not population claims. Evaluation golds are in-distribution to the SFT sources, the scale null is at one (under-fed) data budget, and all blind labels come from LLM (Claude) sub-agents rather than human clinicians — internally consistent (κ .71–.87) but not a substitute for expert human annotation. These are the obvious targets for a follow-up.

Conclusions: discrete > continuous at small scale; collapse is a decode problem; cleaning buys cost, not quality; doubling params at fixed data buys nothing measured; and *blind annotation — here cheap because the annotators are fresh LLM sub-agents — is non-negotiable*: it is what turned our own headline “positive” into a retraction.

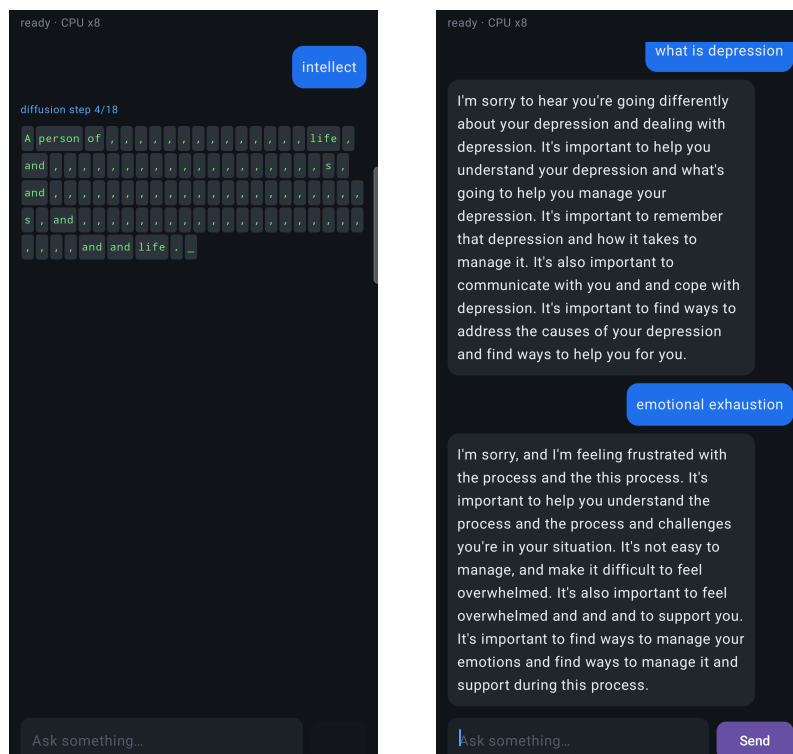


Figure 1: The 66.8M-clean model on a phone (ONNX Runtime Mobile, 8 CPU threads). The two panels are different settings, not a before/after of the same run. Left: masked-diffusion decoding mid-generation under an 18-step preview schedule (step 4/18) — unmasked tokens appear in confidence order. Right: completed chat answers under the frozen evaluation/production §5 sampler (28 steps, temp 0.9, rep 1.3). Outputs are grammatical, on-topic, mildly repetitive — consistent with the limitations stated for this scale.

A Audit map: every headline number, claim $\rightarrow$ artifact $\rightarrow$ status

A closing audit (2026-06-10) re-derived each headline statistic from raw on-disk artifacts rather than from the working notes. Counts were verified against `meta.json` and byte sizes; the blind-trap statistics were recomputed by rerunning `eval/score_blind_traps.py` over the archived key and annotator files; the paired cosines were recomputed end-to-end (re-embedding all outputs) with `eval/paired_sig.py`.

Table 3: Claim $\rightarrow$ artifact $\rightarrow$ recomputed value. Paths relative to `raw_data/004_train/`.

claim	artifact	recomputed
noisy corpus 2.56B tok, 2.16M docs	<code>pretrain_en_diff_clean/meta.json</code>	2,559,832,146; verified
clean corpus 0.774B tok, 833k docs	<code>pretrain_en_clean_v2/meta.json</code>	773,925,936; verified
SFT 261,443 pairs $\times$ 512 params 66.8M / 121.4M	<code>sft_en/meta.json</code> , bin size	exact; verified
runs 20k pre, 8k SFT, $\times$ 6	run logs	verified
decode fix (Table 2)	<code>six mdlm_*.log</code> , 05-29..31	steps and dates verified
retraction p values	<code>RESULTS_matched...md</code>	verified (temp-controlled)
data axis $d = +0.0094$	56-sample key, counts	0.058 / 1.000 / 1.000; verified
scale axis $d = -0.0094$	725 pairs re-embedded	[+0.0003, +0.0186], $t=2.01$, win 0.527; verified
blind scale traps	725 pairs re-embedded	[-0.0188, -0.0001], $t = -1.96$, win 0.479; verified
1000-Q set, gold pairs	<code>annot_{A,B,C}.txt</code> , key	7/28 vs 9/28, $p=0.768$, κ .77/.87/.83; verified
ONNX asset “ $\approx$ 243 MB”	line counts	1000, 725; verified
	app assets dir	286.4 MiB = 300.3 MB; corrected

Archival notes: the §7 blind labels survive as counts in the results file (per-sample files were kept only for the scale round, which fully recomputes); the *noisy* dataset lives in the historically named directory `pretrain_en_diff_clean` (the clean one is `pretrain_en_clean_v2`); Wilson upper bound for 16/28 rounds to 0.73 (notes printed 0.74).

B Blind annotation protocol (standing)

(1) Generate condition pools (here 2 seeds $\times$ 14 traps per checkpoint). (2) Pool, shuffle, strip identifiers; assign neutral IDs (`build_blind_traps*.py`). (3) ≥ 2 (latterly 3) fresh annotators — LLM sub-agents never exposed to source, hypothesis, or direction — label drift yes/no per item; criterion only. (4) Decode by key; report per-source rates, Wilson 95% CIs, Newcombe diff CI, two-sided Fisher exact, pairwise Cohen κ (`score_blind_traps.py`). (5) A non-blind label may motivate, never headline. Triggered: the author 57% vs 29% became blind $p = 1.0$.

C Verbatim prompts given to the LLM sub-agents (Claude)

All annotation and curation labor besides the author was performed by fresh Claude sub-agents (Claude Code **Agent** tool, general-purpose, one task per agent, no memory of the project, no access to hypotheses or checkpoint identities).

Blind drift annotators (§7, §9). Each annotator received only the pooled, shuffled, source-stripped sample file (`blind_traps*.txt`); its header is the entire task specification. Scale round,

verbatim (the §7 round was identical except the domain list lacked physics/cosmology and economics):

Label each item DRIFT=yes/no. DRIFT=yes if a substantial part of the response is about a NON-psychology domain (physics/cosmology, marine biology, automotive, military, pure statistics/math, programming, geometry, economics, anatomy-as-physiology, etc). DRIFT=no if it stays within psychology/therapy/cognition/development EVEN IF factually wrong, vague, or repetitive. Repetition or incoherence alone is NOT drift.

Annotators returned one Sxxx: yes/no line per item (files annot_A-C.txt); decoding against the key happened only afterwards.

1000-question curators (§8–9). Candidate questions were machine-extracted in-distribution from the three SFT sources, psych-filtered, deduplicated, and split into 20 shards of 150; each of 20 sub-agents was instructed to curate, from its shard, the best 50 distinct well-formed English psychology questions, with a further 8 agents reconciling overlaps to exactly 1000; gold answers were recovered for 725 by matching SFT targets. Scripts: `build_q_shards.py`, `build_eval_pairs.py`. Curators never saw model outputs, so the prompt set predates and is independent of every model under comparison.

D Run ledger (verified from logs)

run	steps	wall	checkpoint
mdlm_v1 (noisy pre)	20k	05-29 00:47–10:05	mdlm_v1/final
mdlm_sft (noisy)	8k	10:06–13:48	mdlm_sft/final
mdlm_v1_clean	20k	15:30–00:45	mdlm_v1_clean/final
mdlm_sft_clean (prod)	8k	00:49–04:38	mdlm_sft_clean/final
mdlm_120m	20k	05-29–05-30 21:47	mdlm_120m/final
mdlm_sft_120m	8k	21:50–03:39	mdlm_sft_120m/final

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